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Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

FORMATIVE ASSESSMENT

Details	Please Complete this Section
Name	XOLANI COLLEN Malinga
Surname	Malinga
Organisation:	Easec Technocentric Varsity
Unit/Dept:	Electrical Engineering
Facilitator Name:	

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓
Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed	
Training	Classroom training registers	Training Provider Date:		
Self assessment	Assessment contract signed & dated	Learner file Date:		
Assessment contract	Assessment contract signed & dated	Learner file Date:		
Initial meeting	Assessor briefing checklist	Learner file Date:		
	Assessment contract & assessment plan	Learner file Date:		
Formative Assessment	Assessment instruments	Learner file Date:		
Summative Assessment	Assessment instruments	Learner file Date:		
Other Evidence	Research portfolio (if applicable)	Learner file Date:		
Feedback	Feedback Report	Learner file Date:		
Moderation	Moderators report	Learner file Date:		
Judgement	Assessor Summary Report / Moderator report	Learner file Date:		
1 st Reassessment	Assessors summary report / instruments	Learner file Date:		
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:		

Special arrangements for assessment

Place
Language
Resources
Barriers

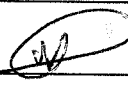
People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

<u>Xolani Malinga</u> 

Learners Name: XOLANI MalingaLearner's signature: 

Date: _____

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

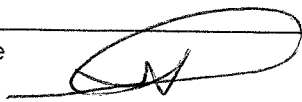
Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: XOLANE MALINGA	Signature: 
Facilitator Name:	Signature:

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Complete the following questions as per instructions provided

LEARNING UNIT 1:**KM-07-KT01: Concepts, theories and principles of supply Systems
(20%)**

Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

Task	Questions Description	Marks
1.	a) Explain the generation and differences between single and three phase alternating current by using wave forms and vector diagrams.	4
	b) Explain by drawing wave form diagrams, the differences between line and phase values.	4
	c) Explain the generation principles of direct current.	4
	d) Calculate power and energy in direct and alternating current circuits	4
	e) Explain with the aid of drawings the operating principles of direct current generators.	4

a) generation and differences Single-phase vs three phase AC
generation of AC (basic principle)

AC is generated using electromagnetic induction

- A Coil rotates in a magnetic field
 - This induces a voltage that changes direction periodically
- Alternating current (AC)

Single-Phase AC

One rotating coil - One sinusoidal Voltage

Output: One waveform

$$y = V_m \sin(\omega t)$$

Waveform: One sin wave

Vector (phasor): One rotating vector

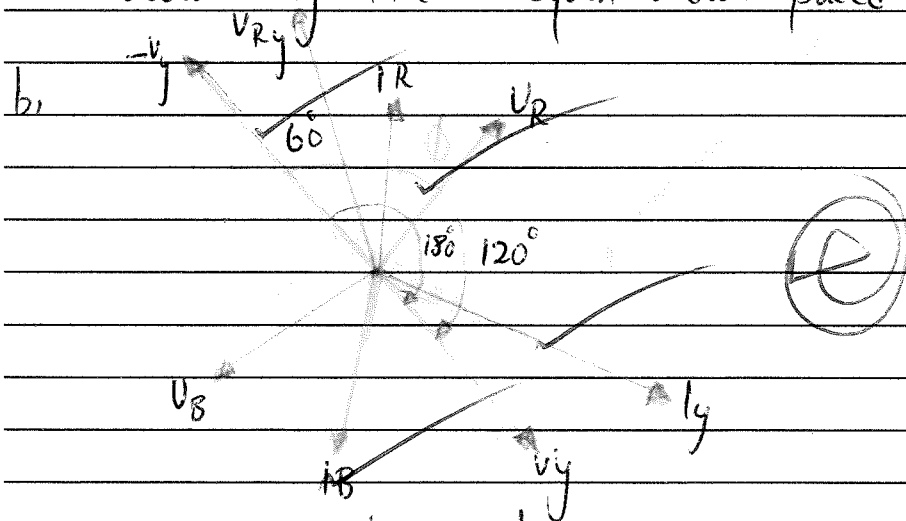
Three-phase AC

- Three coils spaced 120° apart

- produces three sine wave, each shifted by 120°

Waveform: three sine waves (120° apart)

Vector diagram: three equal vectors spaced 120°



c) generation principle of direct Current (DC)

- A rotating coil in a magnetic field (Same as AC)

- But with a commutator instead of Slip rings
- coil rotates $\rightarrow$ induces AC

Commutator reverses connections every half-turn

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c) DC power

$$P = U \times I$$

• AC power

$$P = U \times I \times \cos \theta$$

• $\cos \theta$ - power factor

$$P = UI \cos \theta$$

energy

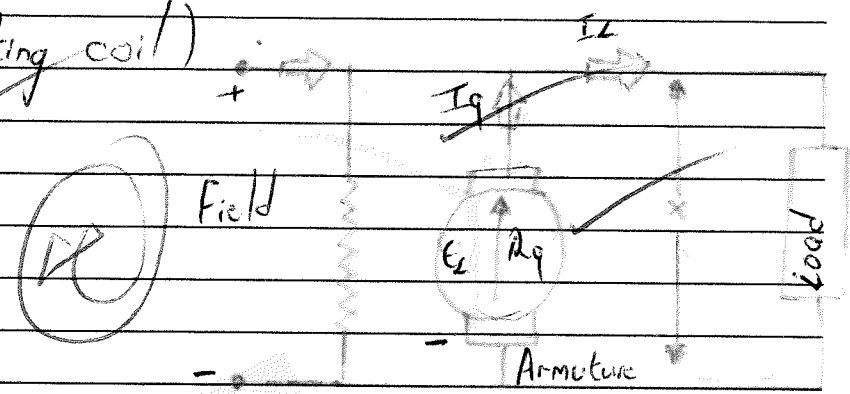
$$E = P \times t$$

DC = Constant Power

AC = depends on power factor and phase angle

e) Main parts.

- Armature (rotating coil)
- field magnets
- Brushes.



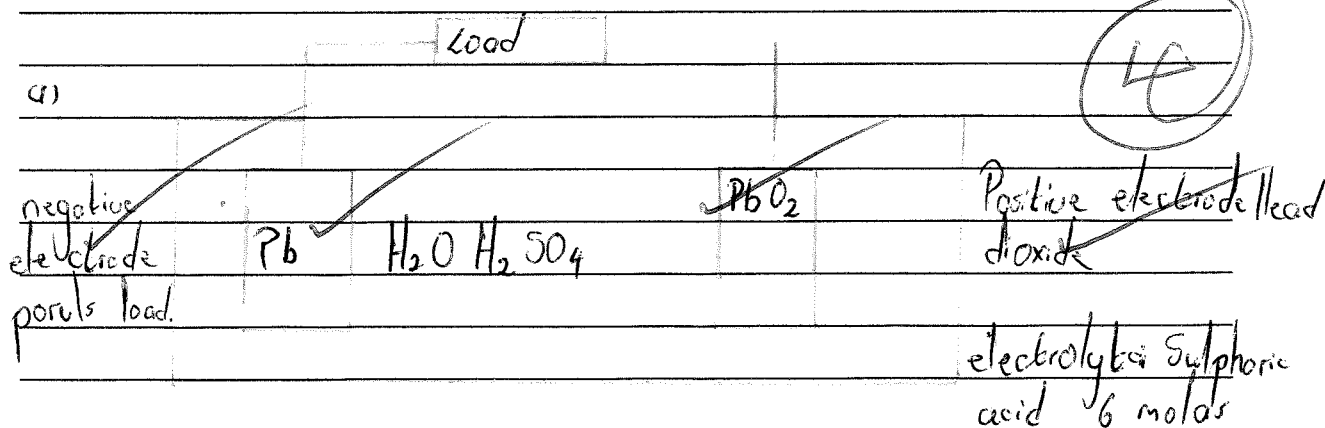
LEARNING UNIT 2:

Batteries

KM-07-KT02: (10%)

Learning Outcome 1:	KT0201 Fundamentals of battery operation
Learning Outcome 2:	KT0202 Types and classification of batteries
Learning Outcome 3:	KT0203 Safety precautions of batteries
Learning Outcome 4:	KT0204 Care and maintenance of batteries
Learning Outcome 5:	KT0205 Battery components and construction
Learning Outcome 6:	KT0206 Disposal of batteries

Task	Questions Description	Marks
2	a) Describe, with the aid of a diagram, the fundamentals of battery construction	4
	b) Describe the advantages and disadvantages of primary and secondary cells for particular applications	4
	c) Name the instruments used for testing batteries, describe how they are used and explain the purpose of each test	4
	d) Explain, with the aid of drawings, the discharging and charging action of lead-acid cells	3



b) Advantages and disadvantages of primary and Secondary

Advantages:

cheap, and simple
ready to use
long shelf life

disadvantages:

Cannot be recharged
Wasteful (Single Use)
limited current output

primary Cells.

Advantages

disadvantages.

High Current output

More expensive initially

Suitable for heavy loads

Requires charging equipment

Reusable (Cost-effective)
Maintenance needed (e.g. lead-acid long term)

c) Voltmeter (Connected across battery terminals)
Measures battery Voltage
Indicates state of charge.

Hydrometer (for lead-acid batteries)

Determines charge level

High density = charged

Low density = discharged.

load tester.

checks battery performance under load

Detects weak or faulty batteries.

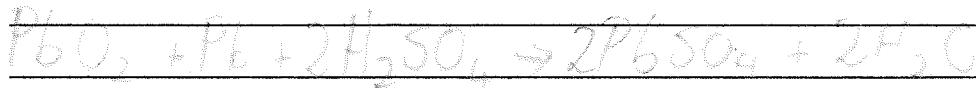
Ammeter (Connected in Series)

Measures charging / discharging current

1. Charging and discharging of lead-Acid Cells (with explanation)

Discharging Process

Battery supplies current to a load.
Chemical reaction occurs



both plates become lead Sulphates ($PbSO_4$)

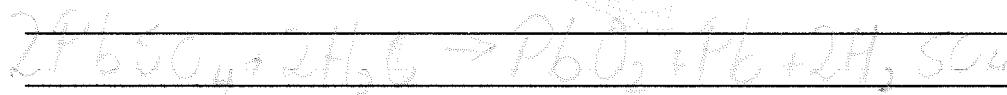
electrolyte becomes weaker

Voltage drops.

Charging Process

External DC Supply is applied
reaction reverse

3

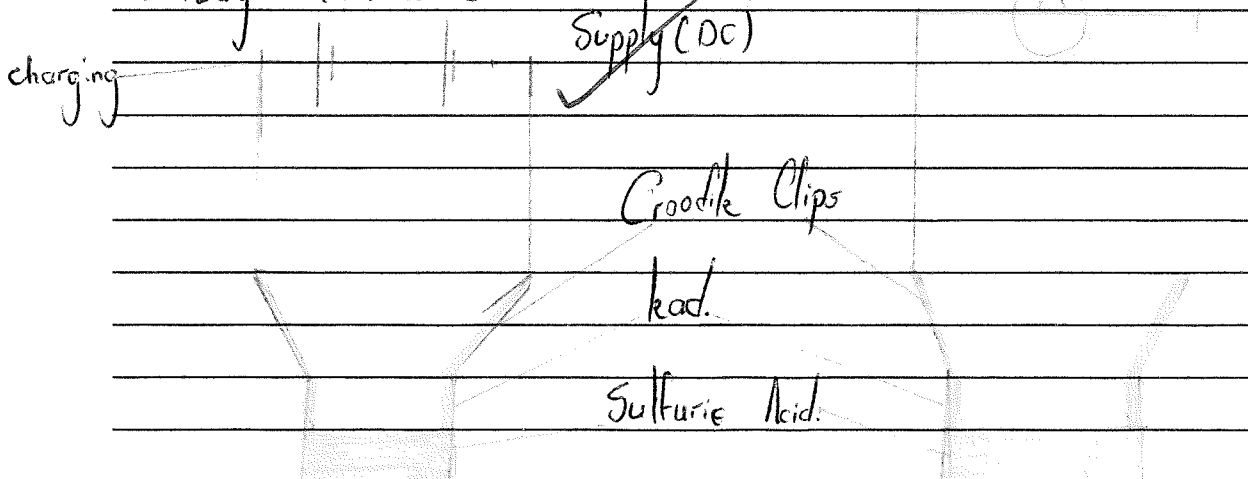


Plates return to original materials

electrolyte becomes stronger

Voltage increases

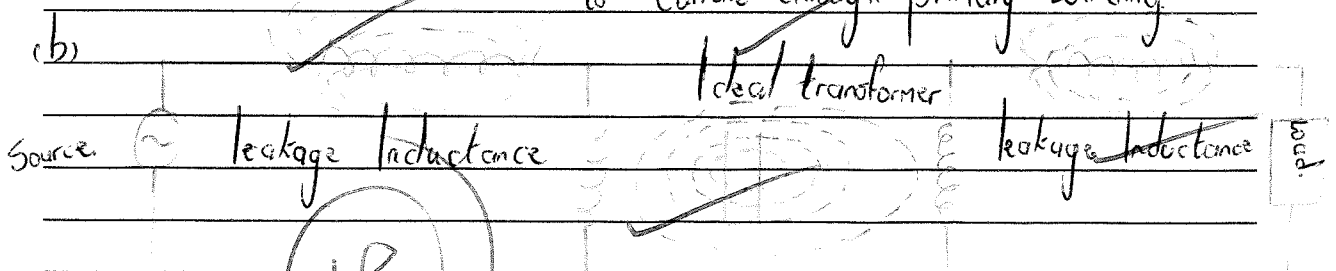
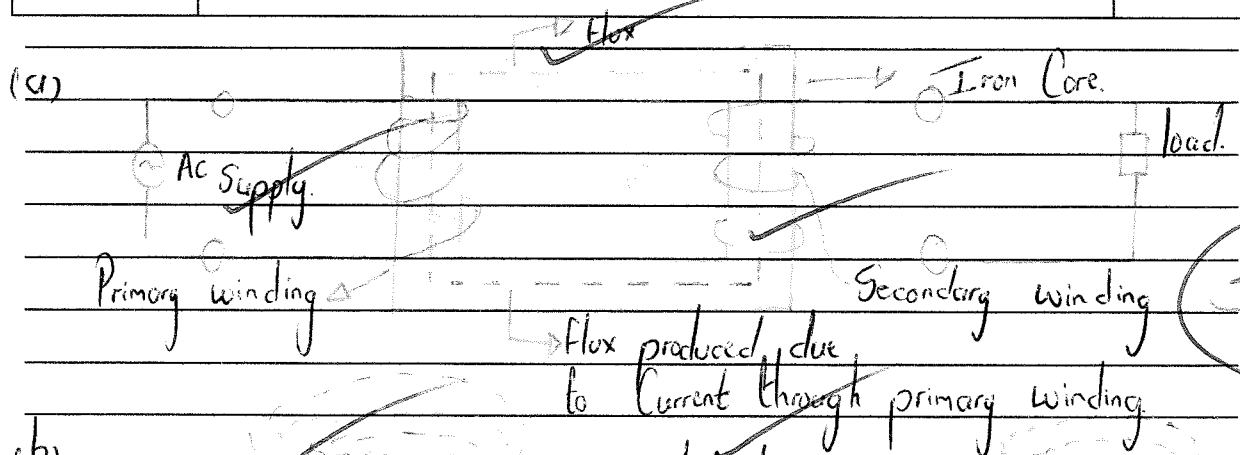
Discharging bulb



LEARNING UNIT 3:**KM-07-KT03: Transformers (20%)**

Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

Task	Questions Description	Marks
3	a) Describe, with the aid of drawings, the construction of single phase transformers	3
	b) Describe, with the aid of drawings, the principle of operation of single phase transformers in terms of mutual inductance and the henry as the unit of inductance	4
	c) Describe types of transformer losses and explain their causes and effects	2
	d) Calculate the terminal voltage, turns and current ratios for an single phase transformer.	3
	e) Describe the various types of construction of three phase transformers - open core, closed core, and shell or divided core	4
	f) Describe the basic transformer maintenance procedures	3
	g) Describe and explain how inductance causes a phase angle between voltage and current and its relationship with the concept 'power factor'	3
	h) Explain the concept of, and need for, transformer tapings	3



c) Core (Iron) losses.

- Hysteresis loss - due to magnetization reversal
- Eddy current loss - Circulating currents in core.

2. Copper losses.

Due to resistance in windings
Depends on current

Other losses

leakage flux

Stray losses

d) Turns Ratio.

$$\frac{N_p}{N_s} = \frac{500}{100} = 5$$

* Terminal Voltage

$$\frac{V_p}{V_s} = \frac{240}{5} = 48V$$

* Primary Current

$$\frac{I_s}{I_p} = \frac{10}{5} = 2A$$

e) types of three-phase transformers.

i. Open Core.

Windings placed on limbs.

Better magnetic path

Shell type.

Core surrounds core

Better magnetic path Better protection and less leakage.

f. Basic transformer maintenance procedures
check oil level (oil-filled transformers)

Inspect for overheating

Tighten connections

Clean bushings and terminals

Test insulation resistance

Check for unusual noise/vibration.

(3)

g) Inductance, phase angle and power factor.

↳ property that opposes change in current.

Phase angle (θ)

Angle between voltage and current

↳ Power factor

(3)

$\cos \theta$ or $\cos \phi$

high inductance $\rightarrow$ large phase angle - low power factor.

h) Transformer Tappings (tap changer)

Changing number of turns to adjust voltage

Maintain constant output voltage

Compensate for supply variations.

(3)

types

↳ Off-load tap changer - adjusted when transformer is off

↳ On-load tap changer (OLTC) - adjusted while operating.

Increasing turns $\rightarrow$ increases voltage

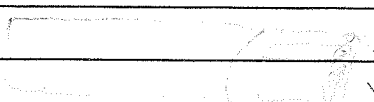
Decreasing turns $\rightarrow$ decreases voltage.

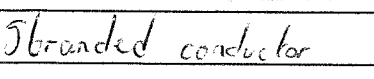
LEARNING UNIT 4:

KM-07-KT04: Types of cables and applications (20%)

Learning Outcome 1:	KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.
Learning Outcome 2:	KT0402 Cable materials and their functions and characteristics
Learning Outcome 3:	KT0403 Identification of cable characteristics and properties
Learning Outcome 4:	KT0404 Applications of various types of cable
Learning Outcome 5:	KT0405 Installation methods and safe use of cables
Learning Outcome 6:	KT0406 Safe transport and storage of cables

Task	Questions Description	Marks
4	a) Describe, with the aid of sketches, the different types of cables and their construction, and explain the function of the various materials used	4
	b) Describe the factors effecting the effeciency of cables	3
	c) Describe the precautions required when transporting and storing cables	3

Insulation  groupe of conductor

Stranded conductor 

C19 Strand  Conductor lag 

Secondary Insulation. Round Cable.

primary insulation
Copper Conductor

b) Factors affecting efficiency cables.

- Material type
- Cross-Sectional Area
- Temperature.

* Conductance represents cables ability to allow electric current to flow, so this factor depends on the type of conductor material type of insulation.

Some factors include length, thickness, frequency, installation, shielding and voltage drop.

ix Transportation Precaution

Prevent damage avoid dropping cable drums from high places especially in temperature below 5°C , to prevent insulation cracking.

Loading lay cable flat on vehicles, but do not stack them on their sides (flanges)

Fixing Secure reels with wedges to prevent rolling

Protection Cover: Cover cable with waterproof material to prevent moisture from entering, particularly during rain or snow

LEARNING UNIT 5:

KM-07-KT05: Switchgear and control gear (20%)

Learning Outcome 1:	KT0501 Principles of operation of switchgear and control gear
Learning Outcome 2:	KT0502 Components of switchgear and control gear systems and the application thereof.
Learning Outcome 3:	KT0503 Electrical drawings
Learning Outcome 4:	KT0504 Introduction to Programmable Logic Controllers (PLC's)
Learning Outcome 5:	KT0505 Introduction to Soft starters
Learning Outcome 6:	KT0506 Introduction to Variable speed drives

Task	Questions Description	Marks
5	a) Describe disconnectors, relays, timers and contactors in terms of construction and operating principles with reference to the contacts, operating coils (where applicable) and operating mechanisms	4
	b) Describe the basic principles of operation for programmable logic controllers	3
	c) Describe the basic principles of operation for variable speed drives	3

a) A disconnector is primary insulation for isolation
Often manually operated.
Relays Use a small current in a coil to switch a large current via contactors
Timers introduce a delay in switching
Contactors are essentially heavy duty relays designed.

②

For high power circuit, typically Using electromagnetic Contactor, working principle.

- Once the contactor coil is energized by the PLC output, an electromagnetic field will be created around.

This electromagnetic field causes the 3 magnetic contactors to close.

Each contactor is connected to the power from one of the 3 phase power supply.

b) Operation PLC

Input Scan - the PLC checks the state (On/Off) of connected input devices.

Sensors, switches, buttons, and stores this data memory. During program execution, the PLC runs the user's control logic based on the inputs received.

3

Output Scan - the PLC updates the output devices to match the decisions made in the logic execution phase.

Housekeeping, the PLC checks its own hardware diagnostics and communicates with other devices.

PLC are Used

Robustness designed to withstand heat, dust and vibration.

Flexibility reprogrammable without re-wiring.

Real time-response - fast, deterministic performance unlike standard components.

Industrial communication - Capable of connecting with HMI (Human Machine Interface) and SCADA Systems.

Ques) A Variable Speed drives sit between the electrical supply and the motor.

Power from the electrical supply goes into a drive and then the drive, the regulates the power that is fed to the motor.

Inside the drive the input power runs through a rectifier that converts the coming AC power to DC power.

3 main components of VFD Included.

A rectifier bridge converter

A direct current (DC) Unit

An inverter.

Voltage-Source Inverter (VSI) drives are by far the most types of drives.

VSD controls the speed and torque of an AC motor by converting fixed frequency and voltage input to a variable frequency and voltage output.

LEARNING UNIT 6:

KM-07-KT06: Lighting systems (10%)

Learning Outcome 1:	KT0601 Principles of Illumination
Learning Outcome 2:	KT0602 Types of luminaries and lighting systems

Task	Questions Description	Marks
6	a) List the different types of alternating current motors and describe their construction	2
	b) Describe Characteristics of single phase and three phase motors	4
	c) Describe and explain how tests are conducted on alternating current motors	2
	d) Calculate alternating current motor properties	2

a) Types of AC Motors.

Induction motors.

Synchronous Motors

Induction motors: Stator that contains laminated iron core slots with a 3-phase windings that produces a rotating magnetic field.

Rotor, which rotates in the field.

Synchronous motors: Stator to an induction it produces a rotating magnetic field.

b) Characteristics of Single-Phase motors.

Shaft power of less than 3kW is used, their circuit has two wire through which the current circulates.

they have power of 10hp.

Characteristics of 3Phase.

Energy is transmitted via three phases of alternating current which phase shifted by 120° from each other.

Each phase has its maximum and minimum voltage and they are simultaneous.

c) AC motor test methods.

Visual inspection and Small test

Initial inspection for burnt-out small, terminal box condition.

Winding Resistance Used to measure resistance between each pair of phase

$$d, S = U \times I$$

$$P = U \times I \cos \theta$$

$$Q = U \times I \sin \theta$$

$$\sin \theta = \frac{Q}{S}$$

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

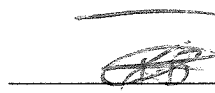
1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1	KT0101 Theories and concepts of alternating current			✓
2	KT0102 Fundamental principles of alternating current			✓
3	KT0103 Alternating current generation (distribution systems theory)			✓
4	KT0601 Principles of Illumination			✓✓
5	KT0602 Types of luminaries and lighting systems			✓✓



Learner Signature

Date



Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT

Learner's Total Mark:

Requirements met ☐

Requirements not met ☐

Action/s required:

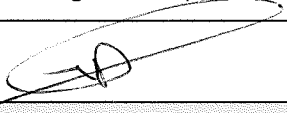
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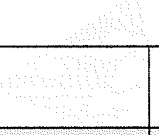
LEARNER FEEDBACK & COMMENTS

Very good Mr Malinga.

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, <u>Xolani Collen Malinga</u>declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR			
I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide

3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments
4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit

